

20260917 - Tutorial (Cdiff eNODE)

September 17, 2026

```
[8]: import numpy as np
import itertools
import pandas as pd

import seaborn as sns
import scipy as sp
from scipy.stats import linregress, pearsonr
import time

import sys
import os

# Get the current working directory (where the script is run from)
current_dir = os.getcwd()

# Get the parent directory
parent_dir = os.path.dirname(current_dir)

# Add the parent directory to sys.path
sys.path.append(parent_dir)

from nsm.enode import *

import matplotlib.pyplot as plt

# set plot parameters
params = {'legend.fontsize': 18,
          'figure.figsize': (8, 7),
          'axes.labelsize': 24,
          'axes.titlesize': 24,
          'axes.linewidth': 3,
          'xtick.labelsize': 20,
          'ytick.labelsize': 20,
          'xtick.labelsize': 20,
          'ytick.labelsize': 20,
          'svg.fonttype': 'none'}
```

```
plt.rcParams.update(params)
np.random.seed(12345)
```

```
[9]: # import data
df = pd.read_csv("../data/sulaiman_data_fmt.csv")
df.head()
```

```
[9]:
```

	Treatments	Time	BT	BV	CH	BU	CS	\
0	BT-BU-BV-CA-CS-DP	0	0.001667	0.001667	0.0	0.001667	0.001667	
1	BT-BU-BV-CA-CS-DP	12	1.116771	0.190538	0.0	0.114889	0.127689	
2	BT-BU-BV-CA-CS-DP	24	1.193564	0.044796	0.0	0.097576	0.297728	
3	BT-BU-BV-CA-DP	0	0.002000	0.002000	0.0	0.002000	0.000000	
4	BT-BU-BV-CA-DP	12	1.116553	0.312622	0.0	0.063024	0.000000	

	CA	DP	DSM	MS001	MS008	MS014
0	0.001667	0.001667	0.0	0.0	0.0	0.0
1	0.002910	0.021037	0.0	0.0	0.0	0.0
2	0.077205	0.031381	0.0	0.0	0.0	0.0
3	0.002000	0.002000	0.0	0.0	0.0	0.0
4	0.001057	0.018078	0.0	0.0	0.0	0.0

```
[10]: # define species and mediator names
species = ['BT', 'BV', 'CH', 'BU', 'CS', 'CA', 'DP', 'DSM', 'MS001', 'MS008', 'MS014']
mediators = []
inputs = []
```

```
[11]: # init model
model = eNODE(dataframe=df,
               species=species,
               mediators=mediators,
               inputs=inputs,
               n_hidden=5, n_layers=2,
               nu2=.001, sigma2=.01)

# init params
model.init_params(sample=False)
print(f"Number of parameters: {model.n_params}")
```

Number of parameters: 156

```
[12]: print(f"Total parameters: {model.n_params}")
t0 = time.time()
model.fit_posterior_EM(lr=.001,
                       n_sample=3,
                       n_sample_hypers=100,
                       max_epochs=10000,
                       patience=1, max_iterations=10)
```

```
print("Elapsed time: {:.3f}s".format(time.time()-t0))
```

```
Total parameters: 156
Updating posterior...
Epoch 0, ELBO: -80502.430, Slope: 1.000
encountered 0 nans
Epoch 10, ELBO: -74543.219, Slope: 1.000
encountered 0 nans
Epoch 20, ELBO: -48105.668, Slope: 0.201
encountered 0 nans
Epoch 30, ELBO: -28056.594, Slope: 0.228
encountered 0 nans
Epoch 40, ELBO: -15644.888, Slope: 0.219
encountered 0 nans
Epoch 50, ELBO: -11915.093, Slope: 0.192
encountered 0 nans
Epoch 60, ELBO: -8119.254, Slope: 0.166
encountered 0 nans
Epoch 70, ELBO: -2221.044, Slope: 0.148
encountered 0 nans
Epoch 80, ELBO: 3649.902, Slope: 0.135
encountered 0 nans
Epoch 90, ELBO: 4975.653, Slope: 0.121
encountered 0 nans
Epoch 100, ELBO: 5193.095, Slope: 0.105
encountered 0 nans
Epoch 110, ELBO: 5226.660, Slope: 0.109
encountered 0 nans
Epoch 120, ELBO: 5253.281, Slope: 0.125
encountered 0 nans
Epoch 130, ELBO: 5272.123, Slope: 0.154
encountered 0 nans
Epoch 140, ELBO: 5350.914, Slope: 0.148
encountered 0 nans
Epoch 150, ELBO: 5512.124, Slope: 0.138
encountered 0 nans
Epoch 160, ELBO: 5701.165, Slope: 0.092
encountered 0 nans
Epoch 170, ELBO: 5910.960, Slope: 0.028
encountered 0 nans
Epoch 180, ELBO: 6051.378, Slope: 0.018
encountered 0 nans
Epoch 190, ELBO: 6166.058, Slope: 0.019
encountered 0 nans
Epoch 200, ELBO: 6250.816, Slope: 0.021
encountered 0 nans
Epoch 210, ELBO: 6317.313, Slope: 0.021
encountered 0 nans
```

Epoch 220, ELBO: 6367.953, Slope: 0.021
encountered 0 nans
Epoch 230, ELBO: 6410.867, Slope: 0.019
encountered 0 nans
Epoch 240, ELBO: 6456.801, Slope: 0.016
encountered 0 nans
Epoch 250, ELBO: 6498.497, Slope: 0.013
encountered 0 nans
Epoch 260, ELBO: 6565.200, Slope: 0.010
encountered 0 nans
Epoch 270, ELBO: 6707.510, Slope: 0.009
encountered 0 nans
Epoch 280, ELBO: 6923.965, Slope: 0.010
encountered 0 nans
Epoch 290, ELBO: 7183.934, Slope: 0.013
encountered 0 nans
Epoch 300, ELBO: 7455.964, Slope: 0.016
encountered 0 nans
Epoch 310, ELBO: 7664.706, Slope: 0.019
encountered 0 nans
Epoch 320, ELBO: 7831.682, Slope: 0.022
encountered 0 nans
Epoch 330, ELBO: 8001.878, Slope: 0.024
encountered 0 nans
Epoch 340, ELBO: 8122.262, Slope: 0.024
encountered 0 nans
Epoch 350, ELBO: 8208.086, Slope: 0.024
encountered 0 nans
Epoch 360, ELBO: 8224.247, Slope: 0.021
encountered 0 nans
Epoch 370, ELBO: 8329.616, Slope: 0.018
encountered 0 nans
Epoch 380, ELBO: 8335.230, Slope: 0.015
encountered 0 nans
Epoch 390, ELBO: 8409.090, Slope: 0.012
encountered 0 nans
Epoch 400, ELBO: 8426.596, Slope: 0.009
encountered 0 nans
Epoch 410, ELBO: 8462.527, Slope: 0.008
encountered 0 nans
Epoch 420, ELBO: 8480.698, Slope: 0.006
encountered 0 nans
Epoch 430, ELBO: 8507.712, Slope: 0.005
encountered 0 nans
Epoch 440, ELBO: 8528.670, Slope: 0.004
encountered 0 nans
Epoch 450, ELBO: 8553.030, Slope: 0.004
encountered 0 nans

Epoch 460, ELBO: 8568.990, Slope: 0.003
encountered 0 nans
Epoch 470, ELBO: 8590.014, Slope: 0.003
encountered 0 nans
Epoch 480, ELBO: 8605.217, Slope: 0.003
encountered 0 nans
Epoch 490, ELBO: 8612.137, Slope: 0.002
encountered 0 nans
Epoch 500, ELBO: 8638.003, Slope: 0.002
encountered 0 nans
Epoch 510, ELBO: 8657.636, Slope: 0.002
encountered 0 nans
Epoch 520, ELBO: 8660.217, Slope: 0.002
encountered 0 nans
Epoch 530, ELBO: 8639.679, Slope: 0.002
encountered 0 nans
Epoch 540, ELBO: 8704.703, Slope: 0.002
encountered 0 nans
Epoch 550, ELBO: 8684.221, Slope: 0.002
encountered 0 nans
Epoch 560, ELBO: 8694.842, Slope: 0.001
encountered 0 nans
Epoch 570, ELBO: 8730.492, Slope: 0.001
encountered 0 nans
Epoch 580, ELBO: 8756.571, Slope: 0.002
encountered 0 nans
Epoch 590, ELBO: 8677.509, Slope: 0.001
encountered 0 nans
Epoch 600, ELBO: 8781.919, Slope: 0.001
encountered 0 nans
Epoch 610, ELBO: 8769.447, Slope: 0.001
encountered 0 nans
Epoch 620, ELBO: 8804.654, Slope: 0.002
encountered 0 nans
Epoch 630, ELBO: 8813.221, Slope: 0.002
encountered 0 nans
Epoch 640, ELBO: 8813.397, Slope: 0.002
encountered 0 nans
Epoch 650, ELBO: 8820.414, Slope: 0.002
encountered 0 nans
Epoch 660, ELBO: 8841.051, Slope: 0.002
encountered 0 nans
Epoch 670, ELBO: 8850.364, Slope: 0.002
encountered 0 nans
Epoch 680, ELBO: 8854.728, Slope: 0.002
encountered 0 nans
Epoch 690, ELBO: 8867.357, Slope: 0.001
encountered 0 nans

Epoch 700, ELBO: 8879.845, Slope: 0.001
encountered 0 nans
Epoch 710, ELBO: 8896.237, Slope: 0.001
encountered 0 nans
Epoch 720, ELBO: 8891.382, Slope: 0.001
encountered 0 nans
Epoch 730, ELBO: 8933.518, Slope: 0.001
encountered 0 nans
Epoch 740, ELBO: 8940.676, Slope: 0.001
encountered 0 nans
Epoch 750, ELBO: 8945.562, Slope: 0.001
encountered 0 nans
Epoch 760, ELBO: 8977.033, Slope: 0.002
encountered 0 nans
Epoch 770, ELBO: 8985.549, Slope: 0.002
encountered 0 nans
Epoch 780, ELBO: 9002.032, Slope: 0.002
encountered 0 nans
Epoch 790, ELBO: 9034.523, Slope: 0.002
encountered 0 nans
Epoch 800, ELBO: 9031.607, Slope: 0.002
encountered 0 nans
Epoch 810, ELBO: 9069.471, Slope: 0.002
encountered 0 nans
Epoch 820, ELBO: 9094.474, Slope: 0.002
encountered 0 nans
Epoch 830, ELBO: 9142.945, Slope: 0.002
encountered 0 nans
Epoch 840, ELBO: 9164.374, Slope: 0.003
encountered 0 nans
Epoch 850, ELBO: 9206.473, Slope: 0.003
encountered 0 nans
Epoch 860, ELBO: 9334.735, Slope: 0.004
encountered 0 nans
Epoch 870, ELBO: 9407.331, Slope: 0.004
encountered 0 nans
Epoch 880, ELBO: 9502.668, Slope: 0.005
encountered 0 nans
Epoch 890, ELBO: 9571.370, Slope: 0.006
encountered 0 nans
Epoch 900, ELBO: 9618.821, Slope: 0.007
encountered 0 nans
Epoch 910, ELBO: 9648.670, Slope: 0.007
encountered 0 nans
Epoch 920, ELBO: 9720.935, Slope: 0.007
encountered 0 nans
Epoch 930, ELBO: 9762.315, Slope: 0.007
encountered 0 nans

Epoch 940, ELBO: 9810.751, Slope: 0.006
encountered 0 nans
Epoch 950, ELBO: 9841.453, Slope: 0.006
encountered 0 nans
Epoch 960, ELBO: 9870.499, Slope: 0.005
encountered 0 nans
Epoch 970, ELBO: 9899.691, Slope: 0.004
encountered 0 nans
Epoch 980, ELBO: 9929.393, Slope: 0.004
encountered 0 nans
Epoch 990, ELBO: 9948.990, Slope: 0.004
encountered 0 nans
Epoch 1000, ELBO: 9967.388, Slope: 0.003
encountered 0 nans
Epoch 1010, ELBO: 9962.562, Slope: 0.003
encountered 0 nans
Epoch 1020, ELBO: 10019.823, Slope: 0.003
encountered 0 nans
Epoch 1030, ELBO: 10032.021, Slope: 0.002
encountered 0 nans
Epoch 1040, ELBO: 10039.046, Slope: 0.002
encountered 0 nans
Epoch 1050, ELBO: 10075.139, Slope: 0.002
encountered 0 nans
Epoch 1060, ELBO: 10092.283, Slope: 0.002
encountered 0 nans
Epoch 1070, ELBO: 10097.730, Slope: 0.002
encountered 0 nans
Epoch 1080, ELBO: 10093.358, Slope: 0.002
encountered 0 nans
Epoch 1090, ELBO: 10140.920, Slope: 0.002
encountered 0 nans
Epoch 1100, ELBO: 10145.600, Slope: 0.002
encountered 0 nans
Epoch 1110, ELBO: 10163.713, Slope: 0.002
encountered 0 nans
Epoch 1120, ELBO: 10177.124, Slope: 0.002
encountered 0 nans
Epoch 1130, ELBO: 10191.552, Slope: 0.002
encountered 0 nans
Epoch 1140, ELBO: 10199.475, Slope: 0.001
encountered 0 nans
Epoch 1150, ELBO: 10209.268, Slope: 0.001
encountered 0 nans
Epoch 1160, ELBO: 10221.949, Slope: 0.001
encountered 0 nans
Epoch 1170, ELBO: 10229.772, Slope: 0.001
encountered 0 nans

Epoch 1180, ELBO: 10237.765, Slope: 0.001
encountered 0 nans
Epoch 1190, ELBO: 10245.317, Slope: 0.001
encountered 0 nans
pass 1
Epoch 1200, ELBO: 10248.930, Slope: 0.001
encountered 0 nans
pass 2
Epoch 1210, ELBO: 10248.860, Slope: 0.001
encountered 0 nans
pass 3
Epoch 1220, ELBO: 10257.195, Slope: 0.001
encountered 0 nans
pass 4
Epoch 1230, ELBO: 10262.450, Slope: 0.001
encountered 0 nans
pass 5
Epoch 1240, ELBO: 10281.403, Slope: 0.001
encountered 0 nans
Updating hyperparameters...
Updating posterior...
Epoch 0, ELBO: 11345.074, Slope: 1.000
encountered 0 nans
Epoch 10, ELBO: 11436.643, Slope: 1.000
encountered 0 nans
Epoch 20, ELBO: 11467.699, Slope: 0.005
encountered 0 nans
Epoch 30, ELBO: 11494.645, Slope: 0.004
encountered 0 nans
Epoch 40, ELBO: 11504.217, Slope: 0.003
encountered 0 nans
Epoch 50, ELBO: 11512.731, Slope: 0.003
encountered 0 nans
Epoch 60, ELBO: 11523.521, Slope: 0.002
encountered 0 nans
Epoch 70, ELBO: 11521.019, Slope: 0.002
encountered 0 nans
Epoch 80, ELBO: 11530.188, Slope: 0.002
encountered 0 nans
Epoch 90, ELBO: 11533.672, Slope: 0.001
encountered 0 nans
pass 1
Epoch 100, ELBO: 11535.827, Slope: 0.001
encountered 0 nans
pass 2
Epoch 110, ELBO: 11544.627, Slope: 0.001
encountered 0 nans
pass 3

Epoch 120, ELBO: 11547.091, Slope: 0.000
encountered 0 nans
pass 4
Epoch 130, ELBO: 11543.606, Slope: 0.000
encountered 0 nans
pass 5
Epoch 140, ELBO: 11546.701, Slope: 0.000
encountered 0 nans
Computing model evidence...

log evidence: 11542.985

Updating hyperparameters...
Updating posterior...
Epoch 0, ELBO: 11923.301, Slope: 1.000
encountered 0 nans
Epoch 10, ELBO: 11936.847, Slope: 1.000
encountered 0 nans
pass 1
Epoch 20, ELBO: 11942.012, Slope: 0.001
encountered 0 nans
pass 2
Epoch 30, ELBO: 11940.831, Slope: 0.000
encountered 0 nans
pass 3
Epoch 40, ELBO: 11944.829, Slope: 0.000
encountered 0 nans
pass 4
Epoch 50, ELBO: 11944.157, Slope: 0.000
encountered 0 nans
pass 5
Epoch 60, ELBO: 11950.044, Slope: 0.000
encountered 0 nans
Computing model evidence...

log evidence: 11945.229

Updating hyperparameters...
Updating posterior...
Epoch 0, ELBO: 12284.672, Slope: 1.000
encountered 0 nans
Epoch 10, ELBO: 12283.647, Slope: 1.000
encountered 0 nans
pass 1
Epoch 20, ELBO: 12282.486, Slope: -0.000
encountered 0 nans
pass 2
Epoch 30, ELBO: 12290.537, Slope: 0.000

encountered 0 nans
pass 3
Epoch 40, ELBO: 12291.776, Slope: 0.000
encountered 0 nans
pass 4
Epoch 50, ELBO: 12289.928, Slope: 0.000
encountered 0 nans
pass 5
Epoch 60, ELBO: 12294.654, Slope: 0.000
encountered 0 nans
Computing model evidence...

log evidence: 12295.476

Updating hyperparameters...
Updating posterior...
Epoch 0, ELBO: 12372.272, Slope: 1.000
encountered 0 nans
Epoch 10, ELBO: 12373.838, Slope: 1.000
encountered 0 nans
pass 1
Epoch 20, ELBO: 12369.299, Slope: -0.000
encountered 0 nans
pass 2
Epoch 30, ELBO: 12370.970, Slope: -0.000
encountered 0 nans
pass 3
Epoch 40, ELBO: 12375.850, Slope: 0.000
encountered 0 nans
pass 4
Epoch 50, ELBO: 12375.181, Slope: 0.000
encountered 0 nans
pass 5
Epoch 60, ELBO: 12374.451, Slope: 0.000
encountered 0 nans
Computing model evidence...

log evidence: 12377.185

Updating hyperparameters...
Updating posterior...
Epoch 0, ELBO: 12346.688, Slope: 1.000
encountered 0 nans
Epoch 10, ELBO: 12348.793, Slope: 1.000
encountered 0 nans
pass 1
Epoch 20, ELBO: 12347.042, Slope: 0.000
encountered 0 nans

```
pass 2
Epoch 30, ELBO: 12350.239, Slope: 0.000
encountered 0 nans
pass 3
Epoch 40, ELBO: 12347.010, Slope: 0.000
encountered 0 nans
pass 4
Epoch 50, ELBO: 12348.171, Slope: 0.000
encountered 0 nans
pass 5
Epoch 60, ELBO: 12346.388, Slope: -0.000
encountered 0 nans
Computing model evidence...
```

```
log evidence: 12352.753
```

```
Updating hyperparameters...
Updating posterior...
Epoch 0, ELBO: 12245.901, Slope: 1.000
encountered 0 nans
Epoch 10, ELBO: 12245.440, Slope: 1.000
encountered 0 nans
pass 1
Epoch 20, ELBO: 12244.646, Slope: -0.000
encountered 0 nans
pass 2
Epoch 30, ELBO: 12245.615, Slope: -0.000
encountered 0 nans
pass 3
Epoch 40, ELBO: 12249.622, Slope: 0.000
encountered 0 nans
pass 4
Epoch 50, ELBO: 12247.765, Slope: 0.000
encountered 0 nans
pass 5
Epoch 60, ELBO: 12250.422, Slope: 0.000
encountered 0 nans
Computing model evidence...
```

```
log evidence: 12245.836
```

```
Elapsed time: 4969.001s
```

```
[13]: model.nu2
```

```
[13]: array([0.00409564, 0.00347636, 0.00348427, 0.00216841, 0.00383533,
           0.00653603, 0.00246703, 0.0061009 , 0.0001    , 0.00877004,
```

```
0.00527298])
```

```
[14]: np.sqrt(model.nu2)
```

```
[14]: array([0.06399719, 0.05896069, 0.05902766, 0.04656617, 0.06193007,  
          0.08084571, 0.04966919, 0.07810823, 0.01          , 0.09364852,  
          0.07261526])
```

```
[15]: model.sigma2
```

```
[15]: array([0.03550557, 0.06213098, 0.10173219, 0.0582415 , 0.38640159,  
          0.22417629, 0.11361313, 0.06699125, 0.06755395, 0.02475161,  
          0.04476646])
```

```
[16]: # Define function to make predictions on test data  
def plot_sample(model, df_test):  
  
    # save true values and predictions  
    true = []  
    pred = []  
    spcs = []  
  
    # pull a random community trajectory  
    for treatment, comm_data in df_test.groupby("Treatments"):  
  
        # make sure comm_data is sorted in chronological order  
        comm_data.sort_values(by='Time', ascending=True, inplace=True)  
        tspan = comm_data.Time.values  
  
        # pull just the community data  
        species_true = comm_data[species].values  
        mediators_true = comm_data[mediators].values  
        input_vals = comm_data[inputs].values[0]  
  
        output_true = np.concatenate((species_true, mediators_true), -1)  
  
        # determine initial condition  
        s0 = species_true[0]  
        r0 = mediators_true[0]  
        x0 = np.concatenate((s0, r0))  
  
        # increase teval  
        t_eval = np.linspace(0, tspan[-1] + 5., 100)  
        steps = len(t_eval)  
  
        # generate posterior samples  
        species_pred, mediators_pred = model.predict_sample(x0,
```

```

t_eval,
n_sample=50,
inputs=input_vals)

# compute summary statistics (mean and variance) of posterior
↳ predictive distribution
E_species_pred, V_species_pred, E_mediators_pred, V_mediators_pred =
↳ model.predict(x0,
↳
↳ t_eval,
↳
↳ n_sample=100,
↳
↳ inputs=input_vals)

# concatenate species and mediator predictions
output_pred = np.concatenate((species_pred, mediators_pred), -1)
E_output_pred = np.concatenate((E_species_pred, E_mediators_pred), -1)
V_output_pred = np.concatenate((V_species_pred, V_mediators_pred), -1)

# predict end-point measured values
true.append(species_true[-1])
pred.append(species_pred[-1])
spcs.append(species)

# plot the results
plt.figure(figsize=(5, 5))
for i, sp in enumerate(species):
    out_true = output_true[:,i]
    plt.scatter(tspan, out_true, color='C{}'.format(i), edgecolor='k',
↳ s=100, zorder=2, label=sp)

# plot posterior samples
for out in output_pred:
    plt.plot(t_eval, out[:, i], color='C{}'.format(i), alpha=.15,
↳ zorder=0)

# plot posterior predictive as a Gaussian
plt.fill_between(t_eval,
E_output_pred[:,i] - V_output_pred[:, i] ** .5,
E_output_pred[:,i] + V_output_pred[:, i] ** .5,
color='C{}'.format(i), alpha=0.2)

# plot mean of posterior
plt.plot(t_eval, E_output_pred[:,i], color='C{}'.format(i),
↳ linewidth=3, zorder=1)

```

```

plt.axhline(y=0, linestyle='--', color='k', alpha=.5)
plt.xlabel("Time (hr)")
plt.ylabel("Species abundance")
# plt.title(input_vals)

# plt.ylim([0, .5])
plt.tight_layout()
plt.show()

```

```
[17]: # plot_sample(model, df)
```

```

[18]: # function to make predictions on test set
def predict_df(model, df):

    # save measured and predicted values
    pred_dfs = []

    for exp_name, exp_df in df.groupby("Treatments"):

        # make sure comm_data is sorted in chronological order
        exp_df.sort_values(by='Time', ascending=True, inplace=True)

        # get initial condition and evaluation times
        t_span = exp_df.Time.values
        Y_m = exp_df[list(species) + list(mediators)].values
        inputs_true = exp_df[inputs].values

        # predict
        s_pred, s_var, m_pred, m_var = model.predict(Y_m, t_span,
↳ inputs=inputs_true[0])

        # save dataframe with predictions
        pred_df = pd.DataFrame()
        pred_df["Treatments"] = [exp_name] * len(t_span)
        pred_df['Time'] = t_span

        # save mean
        pred_df[[f'E[{s}]' for s in species]] = s_pred
        pred_df[[f'E[{m}]' for m in mediators]] = m_pred

        # save variance
        pred_df[[f'V[{s}]' for s in species]] = s_var
        pred_df[[f'V[{m}]' for m in mediators]] = m_var

    pred_dfs.append(pred_df)

```

```
# concatenate list
pred_df = pd.concat(pred_dfs)

return pred_df
```

```
[19]: pred_df = predict_df(model, df)
```

```
[20]: pred_df
```

```
[20]:
```

	Treatments	Time	E[BT]	E[BV]	E[CH]	E[BU]	E[CS]	\
0	BT-BU-BV-CA-CS-DP	0	0.001667	0.001667	0.0	0.001667	0.001667	
1	BT-BU-BV-CA-CS-DP	12	0.867908	0.220758	0.0	0.285090	0.064280	
2	BT-BU-BV-CA-CS-DP	24	1.091385	0.186243	0.0	0.130400	0.210404	
0	BT-BU-BV-CA-DP	0	0.002000	0.002000	0.0	0.002000	0.000000	
1	BT-BU-BV-CA-DP	12	0.891664	0.229696	0.0	0.293501	0.000000	
..	...	...	...	...	...	...	...	
4	MonocultureMS014_0	12	0.000000	0.000000	0.0	0.000000	0.000000	
5	MonocultureMS014_0	15	0.000000	0.000000	0.0	0.000000	0.000000	
6	MonocultureMS014_0	18	0.000000	0.000000	0.0	0.000000	0.000000	
7	MonocultureMS014_0	21	0.000000	0.000000	0.0	0.000000	0.000000	
8	MonocultureMS014_0	24	0.000000	0.000000	0.0	0.000000	0.000000	

	E[CA]	E[DP]	E[DSM]	...	V[BV]	V[CH]	V[BU]	V[CS]	\
0	0.001667	0.001667	0.0	...	0.003477	0.003484	0.002169	0.003836	
1	0.017739	0.036676	0.0	...	0.006858	0.003484	0.007284	0.005479	
2	0.056754	0.034292	0.0	...	0.006495	0.003484	0.003570	0.022094	
0	0.002000	0.002000	0.0	...	0.003477	0.003484	0.002169	0.003835	
1	0.020729	0.040022	0.0	...	0.007320	0.003484	0.007562	0.003835	
..	...	...	...	...	...	...	...	...	
4	0.000000	0.000000	0.0	...	0.003476	0.003484	0.002168	0.003835	
5	0.000000	0.000000	0.0	...	0.003476	0.003484	0.002168	0.003835	
6	0.000000	0.000000	0.0	...	0.003476	0.003484	0.002168	0.003835	
7	0.000000	0.000000	0.0	...	0.003476	0.003484	0.002168	0.003835	
8	0.000000	0.000000	0.0	...	0.003476	0.003484	0.002168	0.003835	

	V[CA]	V[DP]	V[DSM]	V[MS001]	V[MS008]	V[MS014]
0	0.006537	0.002467	0.006101	0.0001	0.00877	0.005273
1	0.006610	0.002625	0.006101	0.0001	0.00877	0.005273
2	0.007436	0.002618	0.006101	0.0001	0.00877	0.005273
0	0.006537	0.002467	0.006101	0.0001	0.00877	0.005273
1	0.006636	0.002654	0.006101	0.0001	0.00877	0.005273
..	...	...	...	...	...	...
4	0.006536	0.002467	0.006101	0.0001	0.00877	0.021761
5	0.006536	0.002467	0.006101	0.0001	0.00877	0.022422
6	0.006536	0.002467	0.006101	0.0001	0.00877	0.022695
7	0.006536	0.002467	0.006101	0.0001	0.00877	0.022815
8	0.006536	0.002467	0.006101	0.0001	0.00877	0.022870

[626 rows x 24 columns]

```
[27]: correlation_list = []

# plot fit
for variable in species:

    measured_values = df.iloc[df.Time.values>0][variable].values
    predicted_values = pred_df.iloc[pred_df.Time.values>0][f'E[{variable}]'].
↪values
    predicted_errors = pred_df.iloc[pred_df.Time.values>0][f'V[{variable}]'].
↪values ** .5

    # Plot the parity plot
    plt.errorbar(measured_values, predicted_values, yerr=predicted_errors,
                  fmt='o', color='blue', ecol='gray', markersize=8,
↪markeredgecolor='k', capsize=3)

    # Add line of perfect agreement
    plt.plot([min(measured_values), max(measured_values)],
              [min(measured_values), max(measured_values)],
              linestyle='--', color='red')

    # Calculate Pearson correlation coefficient
    correlation, _ = pearsonr(measured_values, predicted_values)

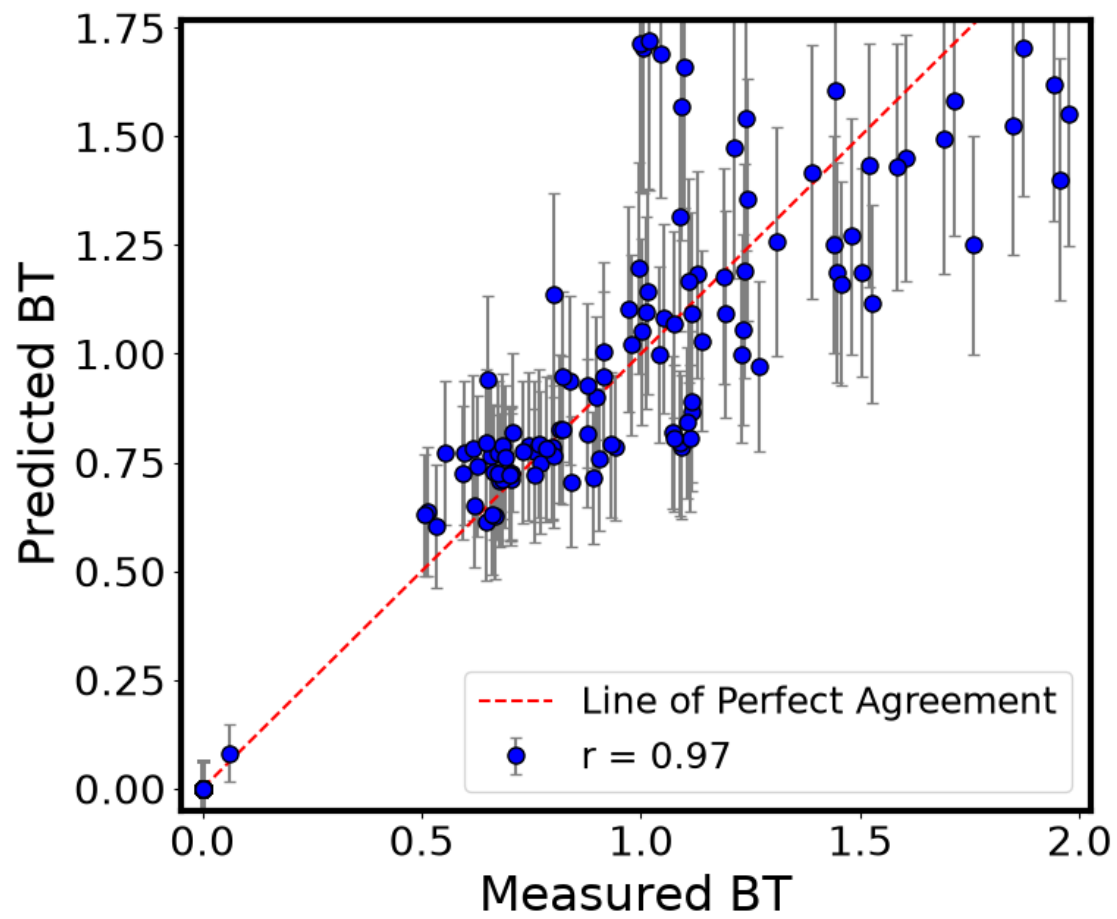
    # Add legend with Pearson correlation coefficient
    plt.legend(['Line of Perfect Agreement', f'r = {correlation:.2f}']) # ,
↪loc='upper left')

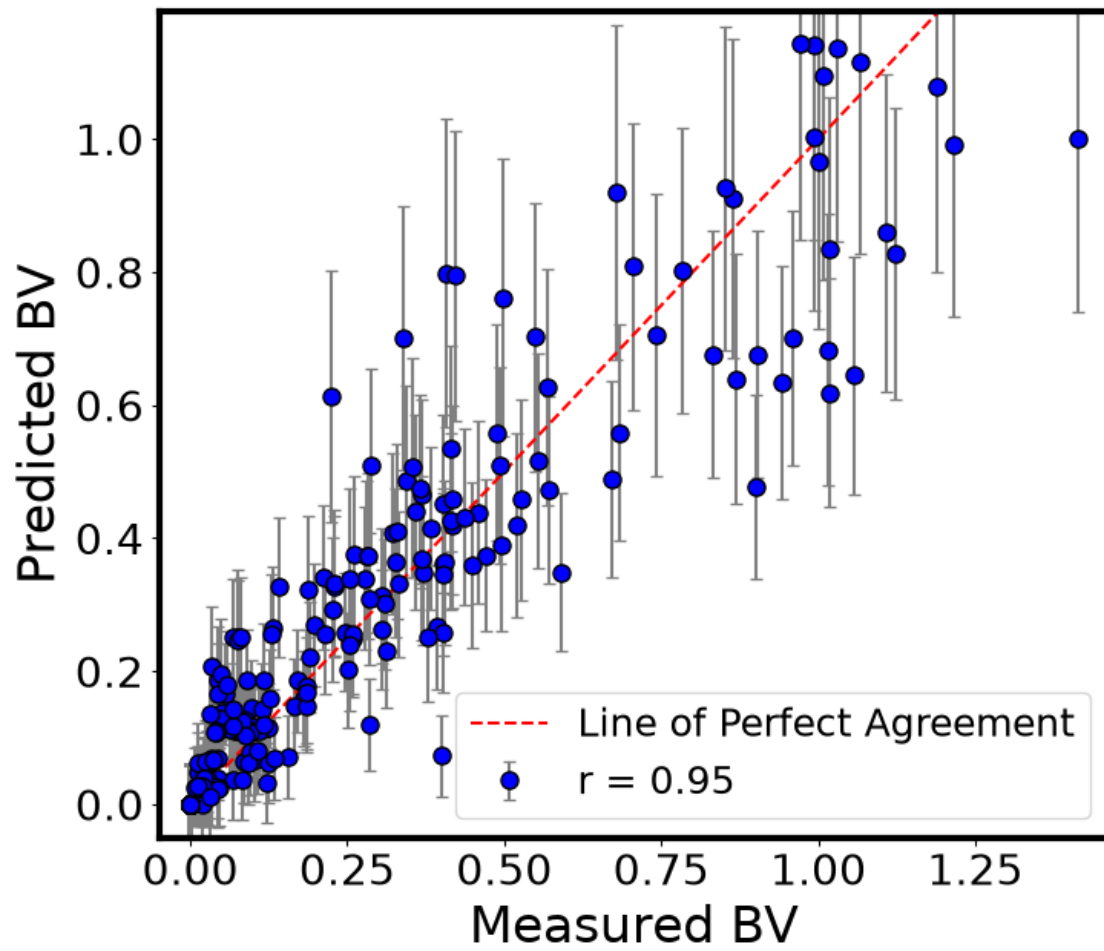
    # Set labels and title
    plt.xlabel(f'Measured {variable}')
    plt.ylabel(f'Predicted {variable}')

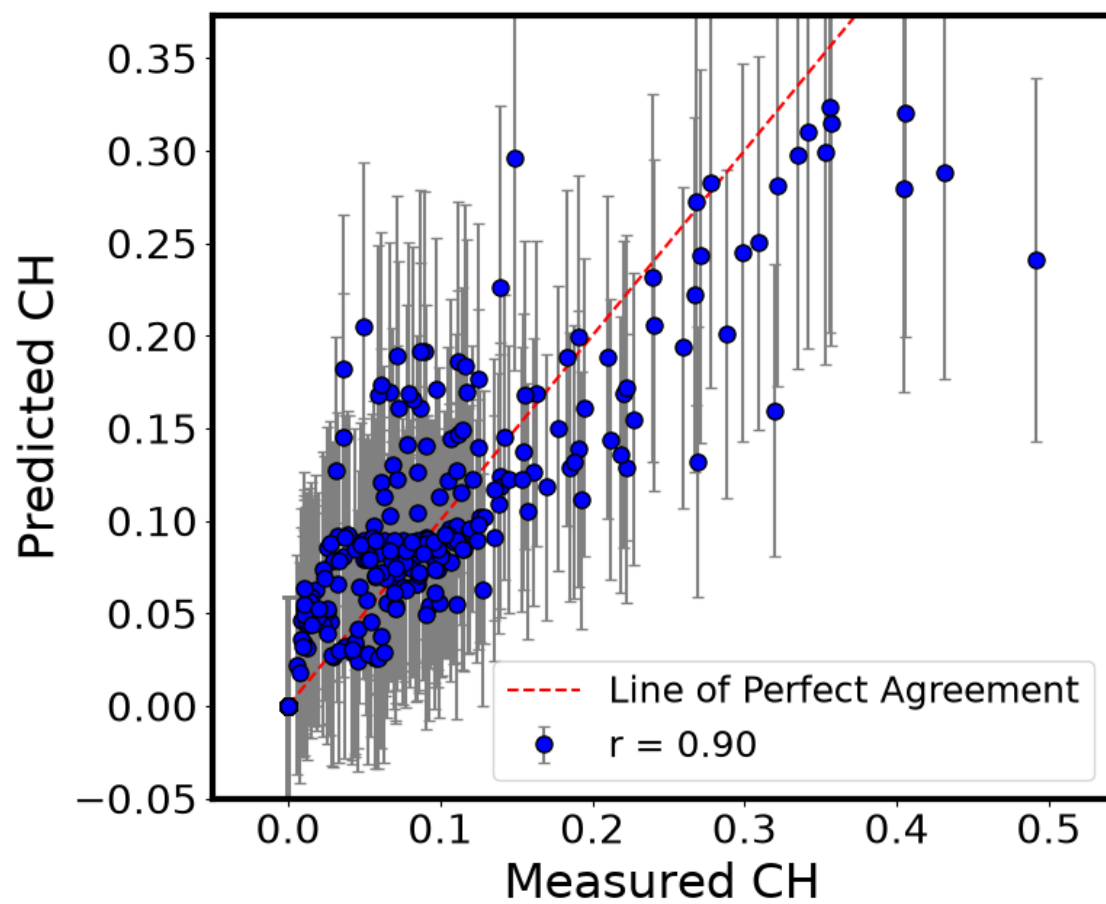
    # Set plot limits
    plt.xlim(min(measured_values)-0.05, max(measured_values)+0.05)
    plt.ylim(min(predicted_values)-0.05, max(predicted_values)+0.05)

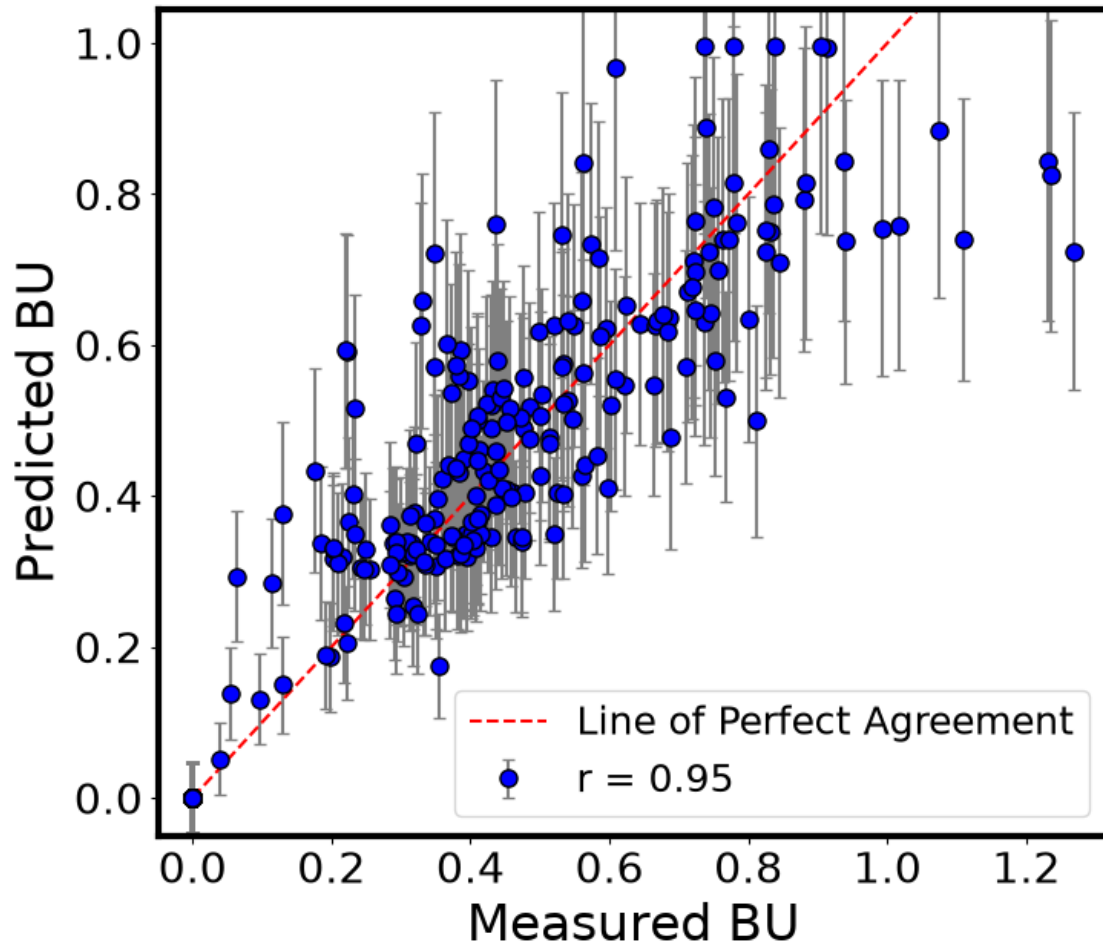
    # Show the plot
    plt.show()

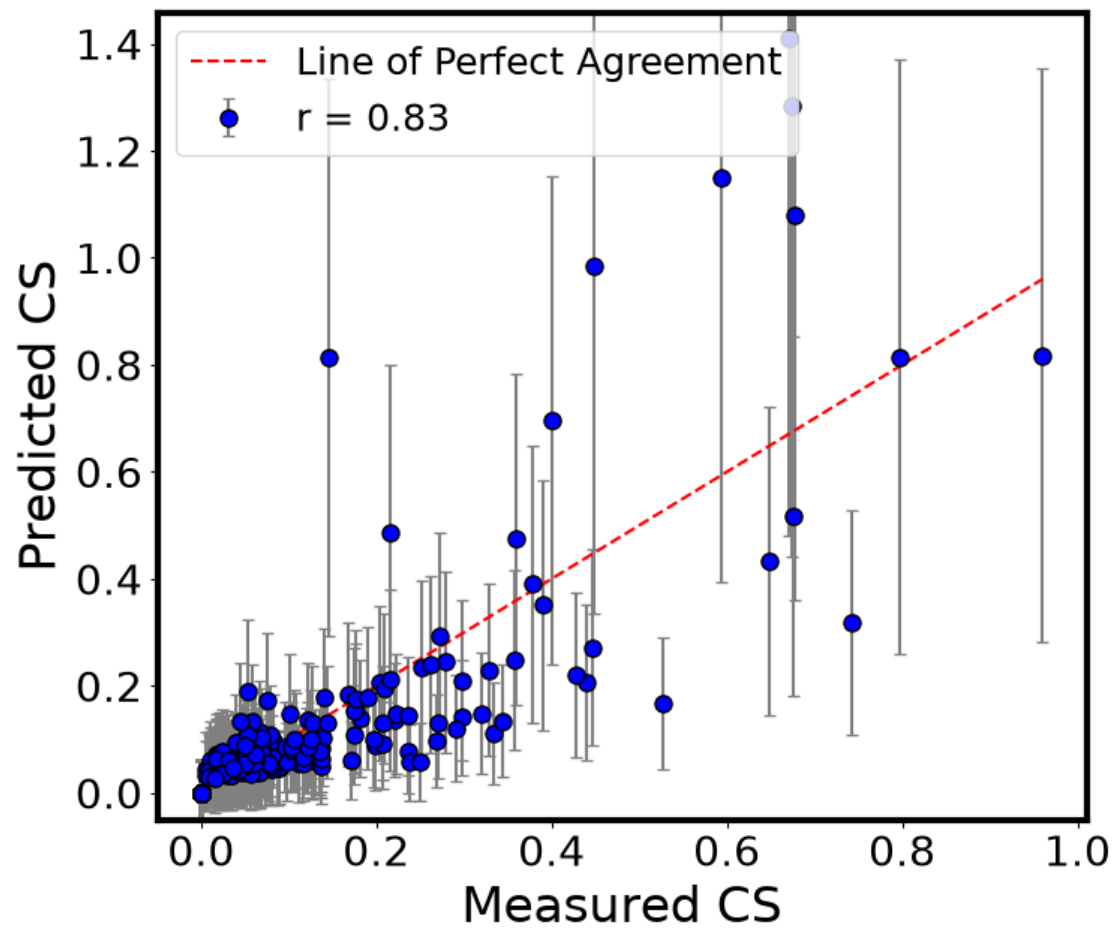
    # get R value
    correlation_list.append(correlation)
```

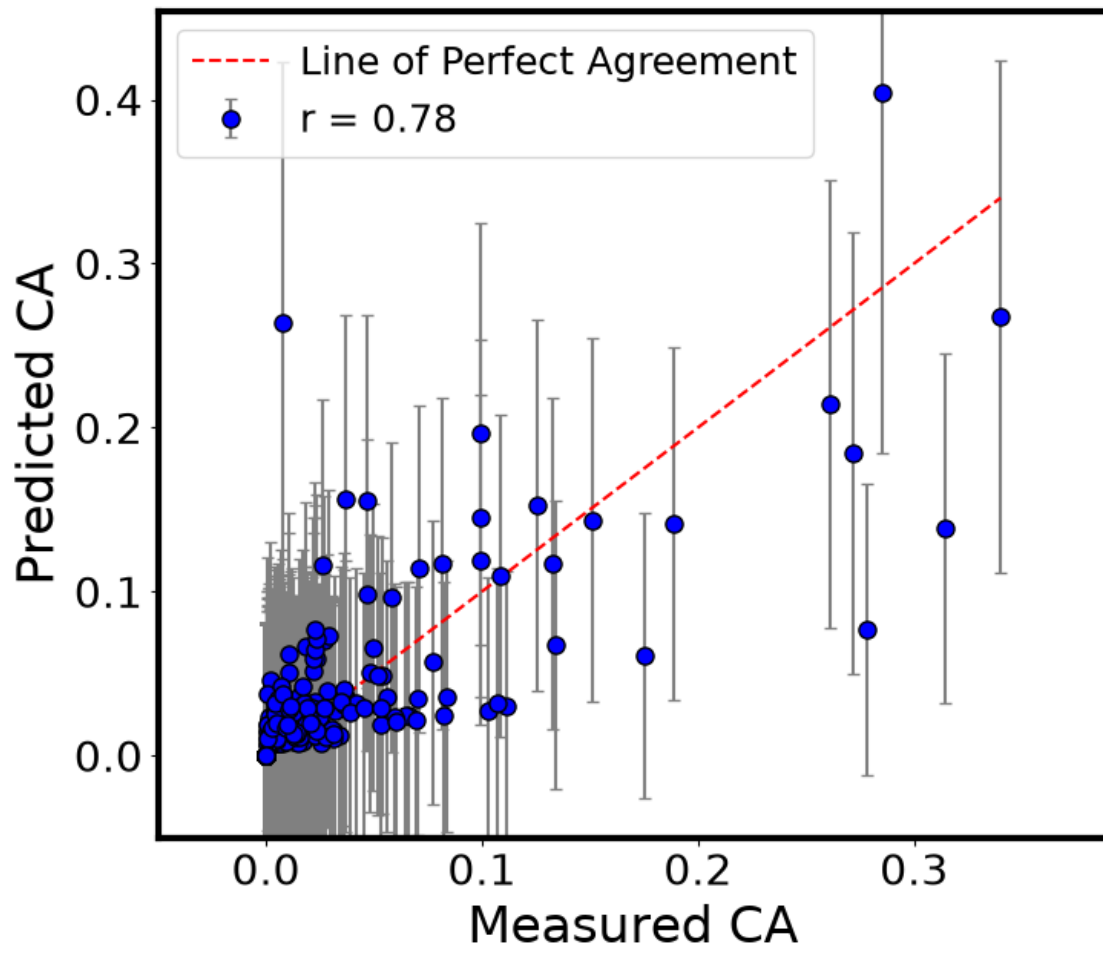


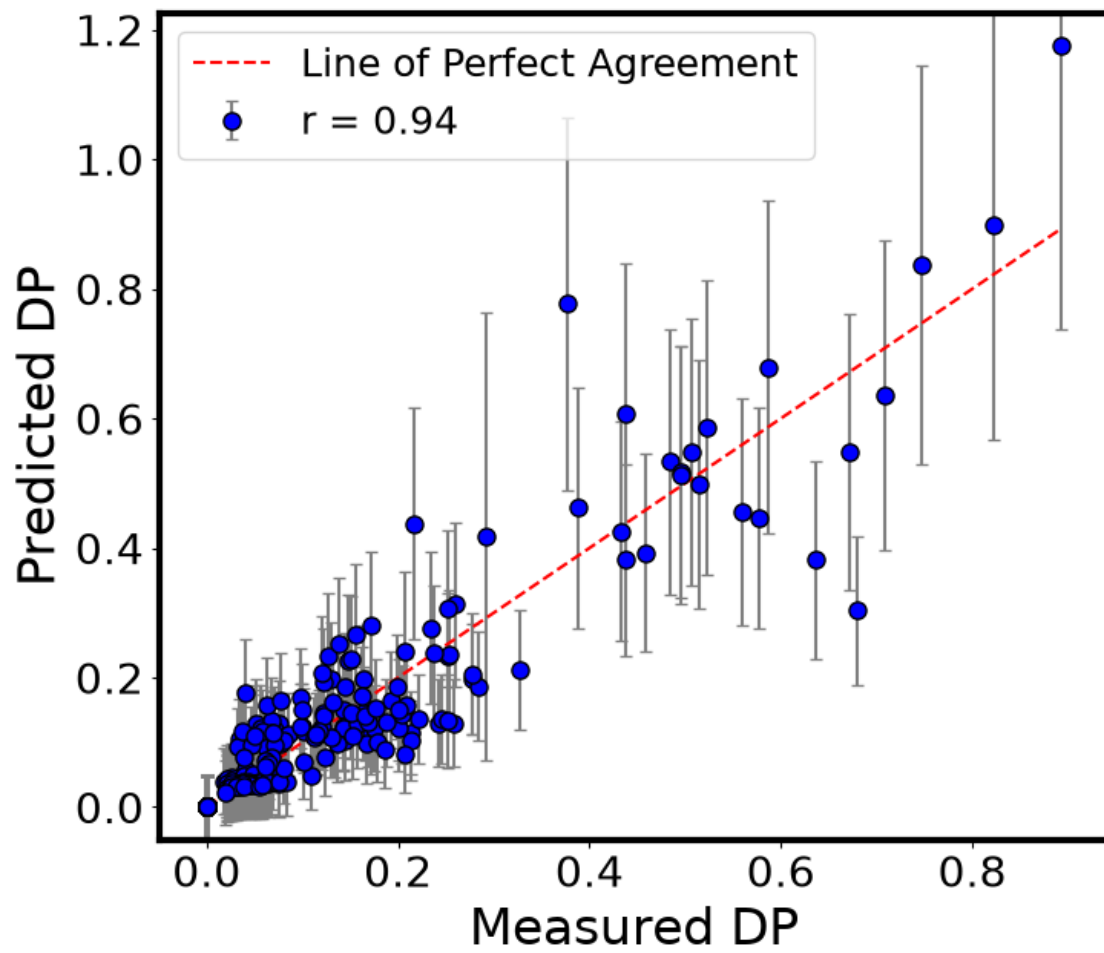


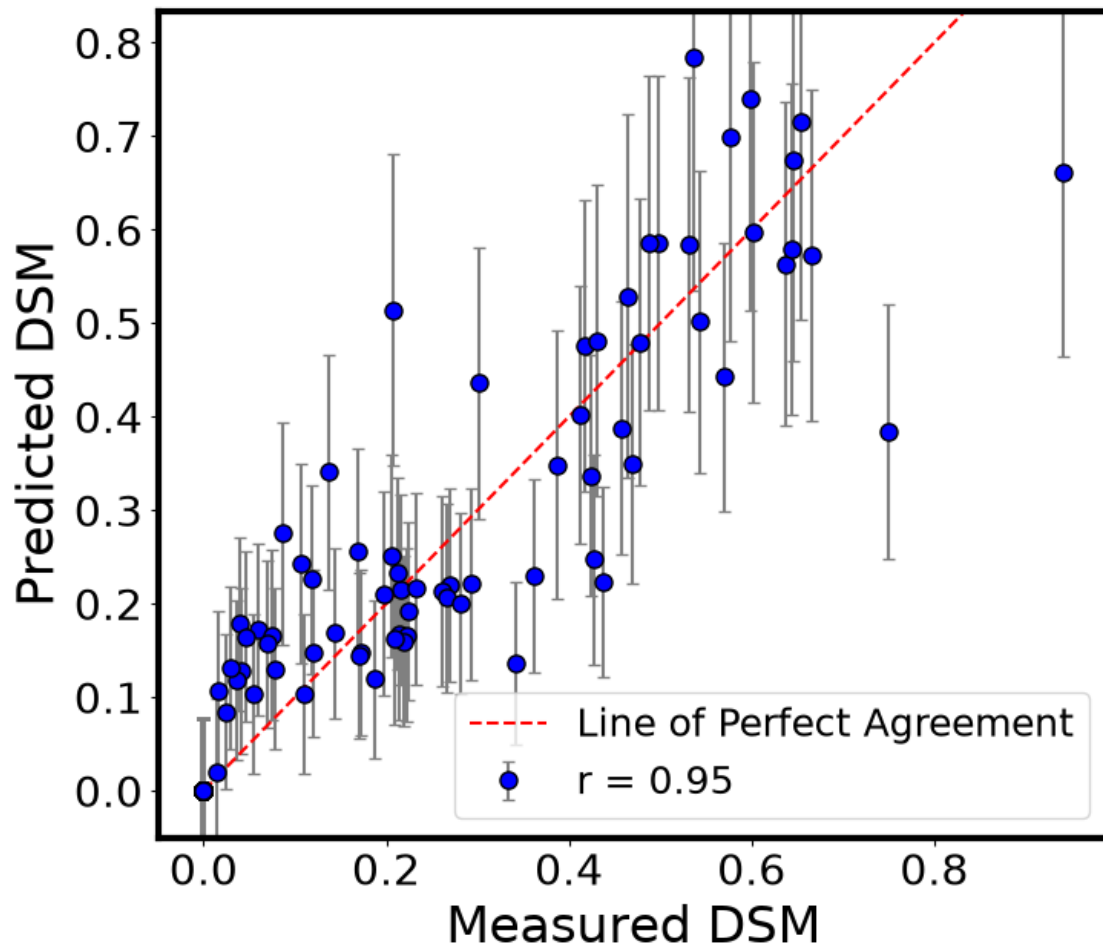


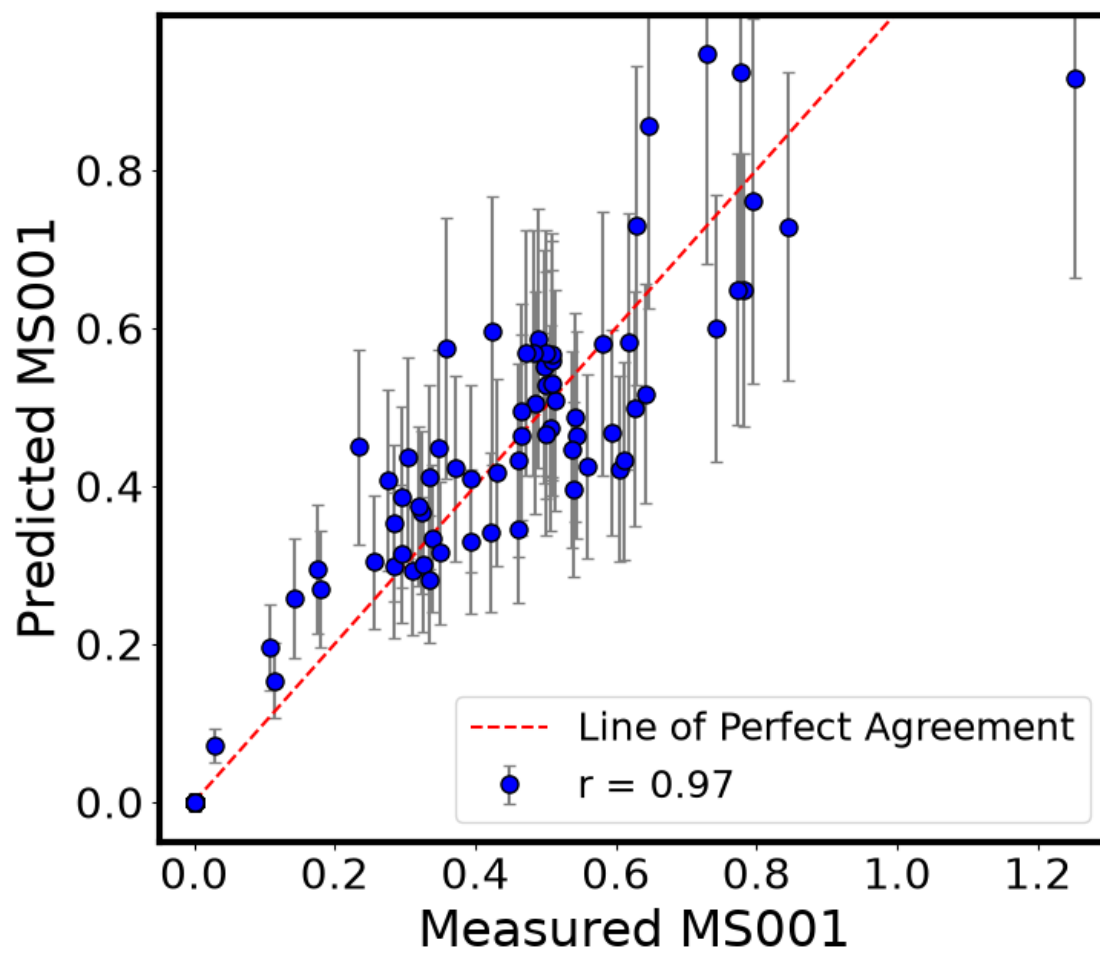


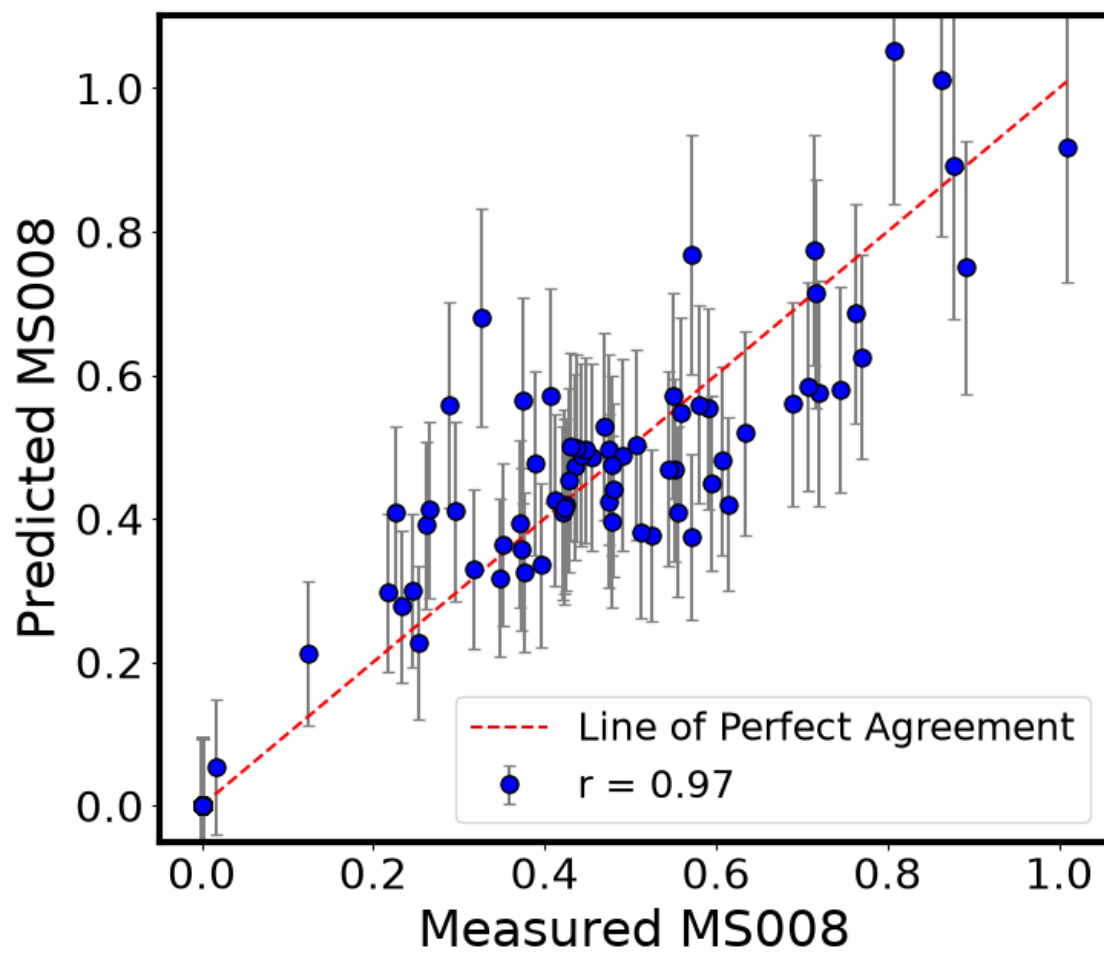


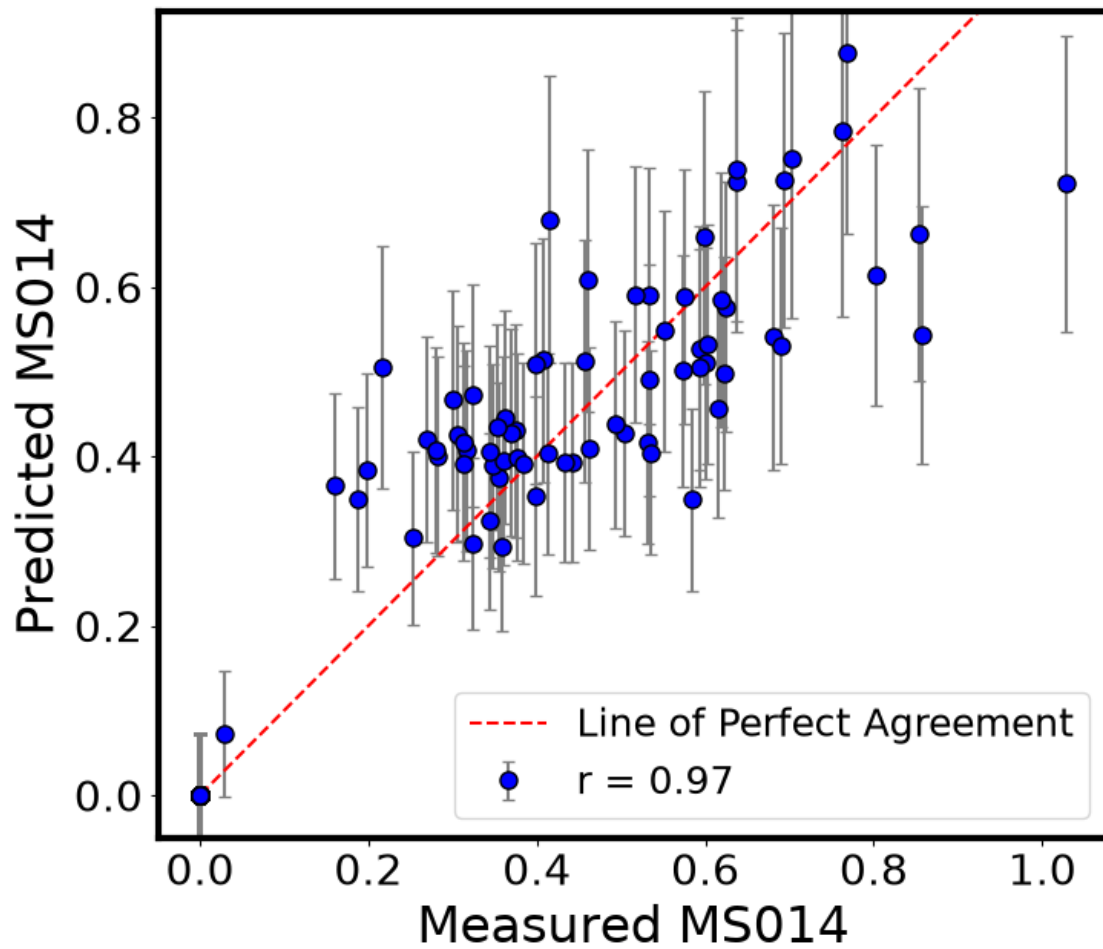












```
[29]: # print(correlation_list)
df_cl = pd.DataFrame({'Value': correlation_list})
df_cl.to_csv('df_cl_2.csv', index=False, float_format='%.4f')
```